

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416895
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Doi; Shigeo

Image forming apparatus including a plurality of doors that are opened and closed

Abstract

An image forming apparatus includes a housing to store an image forming unit that forms an image on a sheet, doors, a cover part, and holding members. Each of the doors rotates around a rotational axis to be opened and closed relative to the housing and includes a positioning shaft and a projection. When each of the doors is in an opened state, an inside of the image forming apparatus is exposed. The cover part is provided in the housing and has fitting holes into which the positioning shafts are fitted. Each of the holding members holds a different one of the doors in a closed state. When each of the doors is in the closed state, each of the positioning shafts included in the doors is fitted into a different one of the fitting holes.

Inventors:	Doi; Shigeo (Ibaraki, JP)
Applicant:	CANON KABUSHIKI KAISHA (Tokyo, JP)
Family ID:	1000008820315
Assignee:	Canon Kabushiki Kaisha (Tokyo, JP)
Appl. No.:	18/629663
Filed:	April 08, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240345532 A1	Oct. 17, 2024

Foreign Application Priority Data

JP	2023-065263	Apr. 12, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: G03G21/00 (20060101); G03G21/16 (20060101)

U.S. Cl.:

CPC G03G21/1633 (20130101); B65H2402/45 (20130101); B65H2405/115 (20130101); G03G2221/1654 (20130101); G03G2221/169 (20130101)

Field of Classification Search

CPC: G03G (21/1633); G03G (2221/169); G03G (2221/1654); B65H (2402/45); B65H (2405/115)

USPC: 399/124

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6374067	12/2001	Park	399/110	E05D 3/022
11281152	12/2021	Kokubu	N/A	G03G 21/1633
2010/0098473	12/2009	Hosoi	271/225	G03G 21/1642
2010/0246109	12/2009	Matsubara	361/679.01	G03G 21/1633
2011/0123217	12/2010	Matsui	399/107	G03G 21/1633
2016/0033928	12/2015	Ishimitsu	399/110	G10K 11/002
2016/0291534	12/2015	Maeda	N/A	G03G 21/1633
2018/0150021	12/2017	Yamamoto	N/A	G03G 21/1609

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2021022584	12/2020	JP	N/A

Primary Examiner: Beatty; Robert B

Attorney, Agent or Firm: Canon U.S.A., Inc. IP Division

Background/Summary

BACKGROUND

Field

(1) The present disclosure relates to an image forming apparatus including a plurality of doors that are opened and closed.

Description of the Related Art

(2) There have been conventionally known a configuration of an image forming apparatus in which an openable and closable door is provided in the housing of the image forming apparatus such that a user can access the inside of the image forming apparatus by opening the door.

(3) Japanese Patent Application Laid-Open No. 2021-22584 discusses an image forming apparatus that includes an openable and closable door that rotates around a rotational shaft and is openable and closable relative to the housing of the image forming apparatus.

(4) As to a door that rotates around a rotational shaft as described above, when the door is closed,

the position of the door relative to the housing of the image forming apparatus may deviate from a design value due to positional misalignment of the door at the time of attachment or deviation or wobbling of a hinge having the rotational shaft. In particular, in an image forming apparatus in which a plurality of doors is arranged side by side in a vertical direction, if the positions of the plurality of doors relative to the housing deviate from the respective design values when the plurality of doors is closed, relative positional misalignment may occur between adjacent doors among the plurality of doors. In the case where the plurality of doors is arranged side by side in the vertical direction in particular, the misalignment between the relative positions of the adjacent doors in a horizontal direction can be easily perceived by users, and the quality of the outer appearance may become deteriorated.

SUMMARY

(5) The present disclosure is directed to, in an image forming apparatus including a plurality of doors, reducing misalignment between relative positions of adjacent doors in a horizontal direction.

(6) According to an aspect of the present disclosure, an image forming apparatus including an image forming unit configured to form an image on a sheet, the image forming apparatus includes a housing configured to store the image forming unit, a plurality of doors, each of which rotates around a rotational axis so as to be opened and closed with respect to the housing and includes a positioning shaft and a projection, wherein, when each of the plurality of doors is in an opened state, an inside of the image forming apparatus is exposed, a cover part that is provided in the housing and has a plurality of fitting holes into which the positioning shafts are fitted, and a plurality of holding members, each of which holds a different one of the plurality of doors in a closed state, wherein, when each of the plurality of doors is in the closed state, each of the positioning shafts included in the plurality of doors is fitted into a different one of the plurality of fitting holes.

(7) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an image forming system.
- (2) FIG. 2 is a schematic cross-sectional view of the image forming system.
- (3) FIGS. 3A to 3C are schematic cross-sectional views of an image forming unit.
- (4) FIG. 4 is a schematic cross-sectional view of a fixing conveyance unit.
- (5) FIGS. 5A and 5B are perspective views of the image forming unit.
- (6) FIGS. 6A and 6B are diagrams illustrating opening and closing of a front door.
- (7) FIGS. 7A and 7B are diagrams illustrating opening and closing of a toner bottle door.
- (8) FIGS. 8A and 8B are diagrams illustrating opening and closing of a collection container door.
- (9) FIG. 9 is a diagram illustrating insertion and removal of a cassette.
- (10) FIG. 10A to 10C are diagrams illustrating an issue with an exterior of an image forming unit in a comparative device.
- (11) FIG. 11 is a diagram illustrating a hinge part biasing unit according to a second exemplary embodiment.
- (12) FIG. 12 is a diagram illustrating opening and closing of a toner bottle door according to the second exemplary embodiment.

DESCRIPTION OF THE EMBODIMENTS

(13) Hereinafter, exemplary embodiments of the present disclosure will be described.

(14) Exemplary embodiments of the present disclosure will be described in a detailed and exemplified manner with reference to the drawings. In the following exemplary embodiments, as

illustrated in FIG. 1, as viewed from an image forming system **100**, the direction toward the front side is defined as frontward direction F, the direction toward the back side (rear side) is defined as backward direction B, the direction toward the left side is defined as leftward direction L, the direction toward the right side is defined as rightward direction R, the direction toward the upper side is defined as upward direction U, and the direction toward the lower side is defined as downward direction D.

(15) However, the dimensions, materials, shapes, and relative arrangements of the components described in the following exemplary embodiments are to be changed as appropriate depending on the configuration of the device to which the present disclosure is applied and various conditions, and they are not intended to limit the scope of the present disclosure thereto.

(16) (Image Forming System)

(17) FIG. 1 is a perspective view of the image forming system **100** according to a first exemplary embodiment. FIG. 2 is a schematic cross-sectional view of the image forming system **100**.

(18) The image forming system **100** includes an image forming apparatus **101**, a large-capacity feeding device **107**, and a sensing device **108**. The image forming apparatus **101** includes an image forming unit **102** and a fixing conveyance unit (fixing unit) **103**. The image forming unit **102** forms a toner image on a recording medium S (hereinafter, called sheet S) fed from the large-capacity feeding device **107** (feeding device) **107**. The fixing conveyance unit **103** fixes the toner image formed by the image forming unit **102** to the sheet S. The image forming unit **102** and the fixing conveyance unit **103** are formed in separate housings. With this configuration, even if the image forming apparatus **101** is of a large size, it is possible to pack and transport the image forming apparatus **101** in a state of being separated into the housings. This improves workability in the physical distribution of the large-sized image forming apparatus **101** from shipment to installation.

(19) Provided on the top of the fixing conveyance unit **103** is an operation unit **104** that is used by an operator to operate the image forming apparatus **101**. The operation unit **104** can be freely changed in position on the upper surface of the image forming system **100**.

(20) The image forming system **100** also includes a document read device **105** and an automatic document feeding device **106** on the top of the image forming unit **102**. The automatic document feeding device **106** feeds documents one by one from a batch of stacked documents to the document read device **105**. The document read device **105** reads the image on the document fed by the automatic document feeding device **106** or the image on the document placed on platen glass (not illustrated).

(21) The document read device **105** and the automatic document feeding device **106** are selectively connected as options to the image forming unit **102**.

(22) The large-capacity feeding device **107** is arranged upstream of the image forming unit **102** with respect to the sheet conveyance direction. The large-capacity feeding device **107** includes a plurality of sheet storage parts. Instead of the large-capacity feeding device **107**, a manual feeding device (not illustrated) or a long-sheet feeding device capable of storing long sheets are selectively connected to the image forming unit **102**. Further upstream of the large-capacity feeding device **107**, another large-capacity feeding device (not illustrated), another manual feeding device (not illustrated), or another long-sheet feeding device (not illustrated) can be selectively connected additionally.

(23) Further, in the image forming system **100** of the present exemplary embodiment, the sensing device **108** is arranged downstream of the fixing conveyance unit **103** with respect to the sheet conveyance direction. However, the sensing device **108** needs not necessarily to be provided but is selectively connected.

(24) In the image forming system **100** of the present exemplary embodiment, post-processing devices not illustrated, such as a discharge tray, a large-capacity stacker, a finisher, and a trimmer, can be selectively connected via the sensing device **108**, and the sheet S on which an image has been formed by the image forming apparatus **101** is fed to a post-processing device that is

connected depending on the purpose.

(25) The sensing device **108** here reads an image on one side or both sides of the sheet S, detects the image density and displacement of the image position, and performs feedback correction to an image signal from the image forming unit **102** based on the detected information.

(26) As described above, when a variety of option devices is selectively connected, the image forming apparatus **101** of the present exemplary embodiment can output products obtained by applying various types of post-processing treatment on various types in an in-line process. This makes it possible to provide the image forming system **100** that achieves high productivity, high image quality, high stability, and high functionality.

(27) (Image Forming Unit)

(28) The image forming unit **102** of the image forming apparatus **101** according to the present exemplary embodiment will be described with reference to FIGS. 3A to 3C.

(29) FIG. 3A is a schematic cross-sectional view of the image forming unit **102**.

(30) The image forming unit **102** includes, as individual image forming units, a yellow image forming station **200Y**, a magenta image forming station **200M**, a cyan image forming station **200C**, and a black image forming station **200K**. The image forming unit **102** also includes laser scanners **203Y**, **203M**, **203C**, and **203K**, toner bottles (toner containers) **205Y**, **205M**, **205C**, and **205K**, and toner supply paths **206Y**, **206M**, **206C**, and **206K** in correspondence with the image forming stations **200Y**, **200M**, **200C** and **200K**, respectively.

(31) Besides those described above, the image forming unit **102** includes primary transfer rollers **207Y**, **207M**, **207C**, and **207K**, an intermediate transfer belt **208**, a toner collection path **210**, collected toner containers **211**, and sheet storage parts **212**.

(32) Further, the image forming unit **102** includes a registration roller **213**, a secondary transfer inner roller **214**, a secondary transfer outer roller **215**, an intermediate transfer belt cleaner **216**, a pre-fixing conveyance belt **217**, and a primary transfer auxiliary roller **218**. The nip portion between the secondary transfer inner roller **214** and the secondary transfer outer roller **215** forms a secondary transfer portion ST.

(33) The yellow image forming station **200Y**, the magenta image forming station **200M**, and the cyan image forming station **200C** have the same structure, which is different from that of the black image forming station **200K**.

(34) FIG. 3B is a diagram illustrating the yellow image forming station **200Y** as a representative of the yellow image forming station **200Y**, the magenta image forming station **200M** and the cyan image forming station **200C**.

(35) The image forming station **200** includes a photoconductive drum **201**, and a primary charger **202**, a development device **204**, and a photoconductive drum cleaner **209** are provided around the photoconductive drum **201**. The primary charger **202** is of a charging roller type.

(36) FIG. 3C is a diagram illustrating the black image forming station **200K**.

(37) Similarly, the black image forming station **200K** includes a photoconductive drum **201K**, and a primary charger **202K**, a development device **204K**, and a photoconductive drum cleaner **209K** are provided around the photoconductive drum **201K**. This station alone also includes a pre-transfer charger **219**. The primary charger **202K** is of a corona charging type.

(38) (Image Forming Process)

(39) The image forming processes in these stations are the same except for the colors of the toners, and thus the image forming process in the yellow image forming station **200Y** will be described as a representative.

(40) Referring to FIG. 3B, the surface of the photoconductive drum **201Y** in the yellow image forming station **200Y** is evenly charged by the primary charger **202Y**. Referring to FIG. 3A, the laser scanner **203Y** emits laser light to the photoconductive drum **201Y** in accordance with image data, thereby forming an electrostatic latent image on the surface of the photoconductive drum **201Y**. The development device **204Y** develops the electrostatic latent image with yellow toner,

thereby forming a yellow toner image on the surface of the photoconductive drum **201Y**. When the toner in the development device **204Y** is consumed by this development, the toner is supplied as appropriate from the tonner bottle **205Y** to the development device **204Y** through the toner supply path **206Y**.

(41) Next, a predetermined pressing force and an electrostatic load bias are applied to the toner image on the surface of the photoconductive drum **201Y** by the primary transfer roller **207Y**, and the toner image on the surface of the photoconductive drum **201Y** is transferred onto the intermediate transfer belt **208**.

(42) After the transfer, the toner slightly left on the photoconductive drum **201Y** is removed by the photoconductive drum cleaner **209Y**. The removed toner is collected by the collected toner containers **211** through the toner collection path **210**.

(43) Similarly, the toner images formed on the other stations are transferred in sequence onto the intermediate transfer belt **208**, and the four toner images are superimposed on top of one another on the belt.

(44) On the other hand, the sheets **S** are fed one by one from one of the sheet storage parts **212** of the image forming unit **102** or from the large-capacity feeding device **107**, and are conveyed to the registration roller **213**. The leading end of each sheet **S** contacts the nip portion of the stopped registration roller **213**, the sheet **S** is formed in a loop shape, and the skew of the sheet **S** is corrected. Thereafter, the registration roller **213** starts to rotate and convey the sheet **S** to the secondary transfer portion **ST** such that the leading end of the toner image on the intermediate transfer belt **208** and the leading end of the sheet **S** are aligned with each other at the secondary transfer portion **ST**. A predetermined pressing force and an electrostatic load bias are applied to the toner image on the intermediate transfer belt **208** at the secondary transfer portion **ST**, and the toner image on the intermediate transfer belt **208** is transferred onto the sheet **S**.

(45) After the transfer, the toner slightly left on the intermediate transfer belt **208** is removed by the intermediate transfer belt cleaner **216**. The removed toner is collected by the collected toner containers **211** through the toner collection path **210**.

(46) The sheet **S** onto which the toner image has been transferred is conveyed to the fixing conveyance unit **103** by the pre-fixing conveyance belt **217**.

(47) (Fixing Conveyance Unit)

(48) FIG. **4** is a schematic cross-sectional view of the fixing conveyance unit **103**.

(49) The fixing conveyance unit **103** includes a fixing unit **301**, a cooling unit **302**, a heat sink **303**, a discharge conveyance path **304**, a discharge reversing unit **305**, a double-sided reversing unit **306**, and a double-sided conveyance path **307**.

(50) The fixing unit **301** heats and presses the toner image on the sheet **S** conveyed from the image forming unit **102** to fix the toner image to the sheet **S**. The heated sheet **S** is cooled by heat absorption by the heat sink **303** arranged in the cooling unit **302**. After that, the sheet **S** is discharged to the sensing device **108** through the ejection conveyance path **304**.

(51) In the case of reversing the front and back of the sheet **S** and discharging the sheet **S**, the sheet **S** is switched back at the discharge reversing unit **305** to switch the leading end and trailing end of the sheet **S**, and the sheet **S** in the front-back reversed state is discharged through the ejection conveyance path **304**.

(52) In the case of forming images on the both sides of the sheet **S**, the sheet **S** with an image formed on a first side of the sheet **S** is switched back at the double-sided reversing unit **306** to switch the leading end and trailing end of the sheet **S**, and the sheet **S** in the front-back reversed state is conveyed to the double-sided conveyance path **307**. After that, the sheet **S** and the subsequent sheet fed from the sheet storage part **212** of the image forming unit **102** or the large-capacity feeding device **107** are conveyed to the registration roller **213** in sequence. Then, an image is formed on a second side of the sheet **S** in the same image forming process as that of the first side, and the sheet **S** is discharged through the discharge conveyance path **304**.

(53) (Monochrome Image Formation)

(54) As described above, the image forming apparatus **101** according to the present exemplary embodiment can form a full-color image using the yellow image forming station **200Y**, the magenta image forming station **200M**, the cyan image forming station **200C**, and the black image forming station **200K**, and can also form a monochrome image using the black image forming station **200K** alone.

(55) In forming a monochrome image, the primary transfer rollers **207Y**, **207M**, and **207C**, the primary transfer auxiliary roller **218**, and the intermediate transfer belt **208** are moved by a separation mechanism (not illustrated) to positions indicated by dotted lines in FIG. **3A**.

Accordingly, the rotational driving of the color image forming stations **200Y**, **200M**, and **200C** separated from the intermediate transfer belt **208** are stopped. That is, in the color image forming stations **200Y**, **200M**, and **200C**, the components constituting the color image forming stations **200Y**, **200M** and **200C** can be prevented from wearing out due to unnecessary rotational driving, and thus the components can achieve longer lives.

(56) Since the photoconductive drum **201K** in the black image forming station **200K** is larger in diameter than the photoconductive drums **201Y**, **201M**, and **201C** in the color image forming stations **200Y**, **200M** and **200C**, the photoconductive drum **201K** is longer in lifetime than the photoconductive drum **201Y**, **201M**, and **201C** in the color image forming stations **200Y**, **200M** and **200C**.

(57) Besides, the primary charger **202K** in the black station is of a contactless corona charging type, which is longer in lifetime than the primary chargers **202Y**, **202M**, and **202C** in the color image forming stations **200Y**, **200M** and **200C** that are of a contact charging roller type.

(58) In addition, the toner bottle **205K** in the black image forming station **200K** has a larger capacity than those of the toner bottles **205Y**, **205M**, and **205C** in the color image forming stations **200Y**, **200M** and **200C**, which is suitable for achieving a longer lifetime.

(59) Accordingly, it is possible to suppress excessive increase in the running cost of the image forming apparatus **101** even if monochrome image formation is executed at a higher ratio than color image formation in the image forming apparatus **101**.

(60) Since the primary charging methods are different between the color image forming stations **200Y**, **200M** and **200C** and the black image forming station **200K** as described above, there is a difference in the wear volume between the color image forming stations **200Y**, **200M** and **200C** and the black image forming station **200K**. As a result, there may occur a difference in toner charging amount between the color photoconductive drums **201Y**, **201M**, and **201C** and the black photoconductive drum **201K**. The difference in toner charging amount may cause an image defect because the toner image is not uniformly transferred onto the sheet **S** in the secondary transfer step. Therefore, the black photoconductive drum **201K** is provided with the pre-transfer charger **219** including a corona charger in order to adjust the toner charging amount to those of the other photoconductive drums.

(61) As described above, according to the configuration of the present exemplary embodiment, it is possible to provide the image forming apparatus **101** that stably achieves high image quality without being significantly affected by the ratio between full-color image formation and monochrome image formation.

(62) (Configuration Outline of Exterior Cover in Housing of Image Forming Unit)

(63) The image forming unit **102** of the image forming apparatus **101** in the present exemplary embodiment is provided with a large number of elements requiring user maintenance. Therefore, as illustrated in FIGS. **5B** and **10A**, the front side of the device is mostly constituted by exterior covers that are all openable and closable.

(64) Specifically, the front side of the device is constituted by a toner bottle door **501** that is opened and closed for replacement of the tonner bottle **205Y**, **205M**, **205C**, or **205K** in which toner is used up, a front door **502** that is opened and closed for removal of jammed sheets or regular

maintenance, a collection container door **503** that is opened and closed for replacement of the collected toner containers **211** that is filled with collected toner, and sheet storage parts **212** that include cassette covers **504** as exterior members to be drawn out in the frontward direction from the device in order to supply sheets.

(65) The above-described three doors are classified into two types of doors that rotate in different rotary directions. The toner bottle door **501** and the collection container door **503** each include a horizontal hinge **512** at the lower end of each door, and are connected to a housing (not illustrated) that stores the image forming unit **102** via the horizontal hinges **512**. Each horizontal hinge **512** includes a rotational shaft extending in the horizontal direction. When the toner bottle door **501** and the collection container door **503** rotate around the respective rotational shafts, the toner bottle door **501** and the collection container door **503** are opened or closed with respect to the housing of the image forming unit **102**. The front door **502** includes a vertical hinge **513** at the right end portion in the horizontal direction, and is connected to the housing via the horizontal hinge **512**. The vertical hinge **513** includes a rotational shaft that extends in the up-down direction that is a vertical direction, and the front door **502** rotates around the rotational shaft with respect to the housing. In the present exemplary embodiment, the vertical hinge **513** is provided at the right end portion of the front door **502** in the front view of the image forming apparatus **101**, but the position of the vertical hinge **513** is not limited to this. That is, the vertical hinge **513** may be provided at the left end portion of the front door **502**. Also, in the toner bottle door **501** and the collection container door **503**, the horizontal hinge **512** may be provided at the upper ends of the doors. Further, in the present exemplary embodiment, the image forming apparatus includes both the door connected by the vertical hinge **513** and the doors connected by the horizontal hinges **512**. However, the present exemplary embodiment is not limited to this, and all the doors may be connected by the vertical hinges **513** or all the doors may be connected by the horizontal hinges **512**.

(66) It is difficult to stably install the doors at designed nominal positions in the closed state, due to deviation or wobbling caused by the configurations of the hinges of the doors and errors in attachment of the doors.

(67) FIG. **10A** illustrates an image forming apparatus that does not include projections **551** to **554** and positioning holes **602** according to the present exemplary embodiment. As described above, the toner bottle door **501** and the collection container door **503** are connected to the housing via the horizontal hinges **512**. The horizontal hinges **512** may wobble in the direction of the rotational shafts of the hinges. Thus, when the toner bottle door **501** and the collection container door **503** are closed, there may occur deviations from the designed nominal positions in the horizontal direction indicated by arrow **1** and arrow **3** in FIG. **10B**.

(68) FIG. **10C** illustrates the state where the position of the closed toner bottle door **501** is deviated from the designed nominal position. As illustrated in FIG. **10C**, when the toner bottle door **501** is shifted in position, there may occur misalignment in the horizontal direction between the relative positions of the toner bottle door **501** and the front door **502** that is adjacent to the toner bottle door **501** in the up-down direction.

(69) On the other hand, the front door **502** is configured to abut against the end in the downward direction when thrust play occurs in the direction of a hinge shaft (not illustrated) of the front door **502**, so that, when the front door **502** is closed, the position of the front door **502** is stabilized in the height direction on the rotation center side. However, the front door **502** tends to incline in a direction indicated by arrow **2** in FIG. **10B** due to poor fitting of the hinge shaft (not illustrated) and the influence of its own weight.

(70) The positional shift caused by the play of the hinge can be reduced by using a high-accuracy hinge, but in that case, there is an issue of cost increase.

(71) The sheet storage parts **212** generally have their own unique positioning configuration for stable sheet feeding performance in the closed state (the sheet feeding-enabled state). In most cases, however, the sheet feeding position is set by adjusting the position of the sheet storage part **212**

itself under the influence of components involving in sheet conveyance from sheet feeding to image transfer, assembly tolerance, and installation environment. Accordingly, it is often difficult to align the relative positions of the cassette cover **504** attached to each sheet storage part **212** and the other exterior covers.

(72) In an image forming apparatus having a plurality of rotary doors as in the image forming apparatus **101** according to the present exemplary embodiment, the positions of the doors **502** to **504** with respect to the housing are set as designed values such that the end portions of the doors **502** to **504** align with one another in the horizontal direction in consideration of the aesthetic appearance. In the present exemplary embodiment, if a misalignment between the relative positions of the end portions of the adjacent doors in the horizontal direction is 2 mm or less, it is defined that the end portions of the doors **502** to **504** align with each other in the horizontal direction. Regulating the misalignment between the positions of the end portions in the horizontal direction to 2 mm or less can give users an impression that the end portions of the doors **502** to **504** in the closed state align with each other in the horizontal direction. However, if the positions of the doors **502** to **504** in the closed state with respect to the housing are deviated from the design values due to the positional shifts described above, the misalignment between the relative positions of the adjacent doors in the horizontal direction may be 2 mm or more.

(73) In a design in which corner portions or edges are prominent in the shapes of the exterior members as in the image forming apparatus **101** according to the present exemplary embodiment, there is an issue that misalignment between the relative positions of the exterior covers is noticeable in particular.

(74) (Positioning Configuration of Openable and Closable Exterior Cover According to Present Invention)

(75) The image forming apparatus **101** according to the present exemplary embodiment includes a left cover **505(a)** on the front portion of the left side surface of the image forming unit **102** as illustrated in FIGS. 5A and 5B. Although described below in detail, a plurality of fitting holes into which a plurality of projections including a projection **552** is fitted is formed on the left cover **505(a)**.

(76) FIG. 6A illustrates the front door **502** in the opened state with respect to the housing, and FIG. 6B illustrates a handgrip **506** of the front door **502** and its vicinity in an enlarged view. Since the front door **502** is provided with the handgrip **506**, the front door **502** can be opened and closed by grabbing the handgrip **506**. The front door **502** is provided with a magnet **510** on the inner side and the housing is provided with a metallic part **509**. When the front door **502** is in the closed state, the magnet **510** attracts the metallic part **509** to hold the front door **502** in the closed state. In order to open the closed front door **502**, it is necessary to exert a force stronger than the force of the magnet **510** attracting the metallic part **509**, so that it is possible to reduce the risk of the closed front door **502** becoming accidentally opened. In the present exemplary embodiment, the attraction force between the magnet **510** and the metallic part **509** is used as a mechanism for holding the front door **502** in the closed state. However, another mechanism may be used as far as it can hold the front door **502** in the closed state. Further, in the present exemplary embodiment, the housing is provided with an open/close sensor **507**, and the front door **502** is provided with an open/close sensor projection **508** on its inner side. The open/close sensor **507** is shaped such that the open/close sensor projection **508** is insertable thereinto. When the front door **502** is in the closed state, the open/close sensor projection **508** is inserted into the open/close sensor **507**, so that the open/close sensor **507** detects that the front door **502** is in the closed state. The open/close sensor projection **508** is inserted into the open/close sensor **507** when the front door **502** is closed, but does not have a function of restricting the position of the closed front door **502** and has a function different from those of projections **551** to **554** and positioning holes **601** described below. The front door **502** is further provided with a projection **552** on the inner side, and the left cover **505(a)** is provided with a positioning hole **601(b)** into which the projection **552** is fitted when the front door

502 is in the closed state. The projection **552** is shaped such that its diameter is smaller as the distance from the front door **502** is larger, and is shaped so as to be guided into the positioning hole **601(b)** when the front door **502** is closed. The projection **552** and the positioning hole **601(b)** are provided such that when the front door **502** is in the closed state and the projection **552** fits into the positioning hole **601(b)**, the position of the front door **502** meets the design value. Therefore, even if there occurs play of the horizontal hinge **512** or a positional shift of the front door **502** at the time of attachment, fitting the projection **552** into the positioning hole **601(b)** makes it possible to set the position of the front door **502** with respect to the housing to the position of the design value.

(77) As illustrated in FIG. **6B**, the projection **552** protrudes in a direction away from the front door **502**, and is tapered such that the cross section area is reduced as the distance from the front door **502** increases. Forming the shape of the projection **552** as described above makes it possible to guide the projection **552** to the positioning hole **601(b)** even if the centers of the projection **552** and the positioning hole **601(b)** are not aligned. The projection **551** and the projection **553** described below have the same shape. The shapes of the projection **552** and the positioning hole **601(b)** are not limited to those described above, and the positioning hole **601(b)** may be tapered such that the opening area is reduced in the insertion direction of the projection **552**. The projections **551** to **553** may not be all identical in shape but may be different in shape. In addition, the projections **551** to **553** may have a shape with a uniform cross section area.

(78) FIGS. **7A** and **7B** illustrate the toner bottle door **501** in the opened state with respect to the housing. The toner bottle door **501** is also provided with the magnet **510** (not illustrated in FIGS. **7A** and **7B**) on the inner side of the toner bottle door **501**. When the toner bottle door **501** is in the closed state, the magnet **510** attracts the metallic part **509** (not illustrated in FIGS. **7A** and **7B**) to hold the toner bottle door **501** in the closed state.

(79) Similarly to the front door **502**, the toner bottle door **501** is provided with the open/close sensor projection **508** (not illustrated in FIGS. **7A** and **7B**). When the toner bottle door **501** is in the closed state, the open/close sensor projection **508** is inserted into the open/close sensor **507** (not illustrated in FIGS. **7A** and **7B**), so that the open/close sensor **507** detects that the toner bottle door **501** is in the closed state. Further, similarly to the front door **502**, the toner bottle door **501** is provided with the projection **551**. When the toner bottle door **501** is in the closed state, the projection **551** fits into a positioning hole **601(a)**. Thus, the position of the toner bottle door **501** with respect to the housing can be set to the position of the design value even if play of the vertical hinge **513** or a positional shift of the toner bottle door **501** at the time of attachment occurs.

(80) FIGS. **8A** and **8B** illustrates the collection container door **503** in the opened state with respect to the housing. The collection container door **503** is also provided with the magnet **510** (not illustrated in FIGS. **8A** and **8B**) on the inner side of the collection container door **503**. When the collection container door **503** is in the closed state, the magnet **510** attracts the metallic part **509** (not illustrated in FIGS. **8A** and **8B**) to hold the collection container door **503** in the closed state. Similarly to the front door **502**, the collection container door **503** is provided with the open/close sensor projection **508** (not illustrated in FIGS. **8A** and **8B**). When the collection container door **503** is in the closed state, the open/close sensor projection **508** is inserted into the open/close sensor **507** (not illustrated in FIGS. **8A** and **8B**), so that the open/close sensor **507** detects that the collection container door **503** is in the closed state. Further, similarly to the front door **502**, the collection container door **503** is provided with the projection **553**. When the collection container door **503** is in the closed state, the projection **553** fits into a positioning hole **601(c)**. Thus, the position of the collection container door **503** with respect to the housing can be set to the position of the design value even if play of the horizontal hinge **512** or a positional shift of the collection container door **503** at the time of attachment occurs.

(81) As described above, when the toner bottle door **501**, the front door **502**, and the collection container door **503** are closed, the projections **551** to **553** fit into the positioning holes **601(a)** to **601(c)**, respectively. Therefore, the positions of the toner bottle door **501**, the front door **502** and

the collection container door **503** in the closed state with respect to the housing are set to the positions of the design values, so that the positions of the end portions of the toner bottle door **501**, the front door **502** and the collection container door **503** can be aligned with one another in the horizontal direction.

(82) FIG. **9** illustrates the cassette covers **504** in the opened state with respect to the housing. In a case where the sheet storage parts **212** have their own unique positioning configuration as described above, the projection **554** of each cassette cover **504** may not be properly inserted into a positioning hole **601(d)** when the cassette is closed. Thus, providing a fixing unit (not illustrated) in each cassette cover **504** that allows the cassette cover **504** to be freely movable by a predetermined amount in the horizontal direction with respect to the sheet storage part **212** makes it possible to achieve smooth fitting of the cassette covers **504** and easy alignment with other doors and exterior parts.

(83) In the present exemplary embodiment, the left cover **505(a)** having the function of positioning the openable and closable exterior covers is provided on the left side surface of the image forming unit **102**. However, the same advantageous effect can be achieved by providing a cover having the same function on the right side surface of the image forming unit **102**.

(84) In addition, in the present exemplary embodiment, the positioning holes **601** formed in the left cover **505(a)** are vertically long holes and are provided to perform horizontal positioning alone. In order to perform positioning of the doors in the height direction, a cover member having the same positioning function as the left cover is provided on the right and left sides, the positioning holes in one of the cover members are made circular, and the positioning holes in the other cover member are made long in the horizontal direction. This configuration makes it possible to properly position the doors.

(85) In the present exemplary embodiment, the positioning holes **601(a)** to **601(c)** provided in the left cover **505(a)** are arranged in a line along the vertical direction. Providing the positioning holes **601** as described above makes it easy to attach the doors **501** to **503** to the housing. The positions of the positioning holes **601** are not limited to the above-described positions, and can be set to any positions where, when the projections **551** to **553** are fitted, the positions of the doors **501** to **503** with respect to the housing meet the positions of the design values.

(86) Further, in the present exemplary embodiment, the projections are provided on the openable and closable exterior covers, and the positioning holes **601** are provided in the left cover **505(a)**. However, the same advantageous effect can be achieved by providing the positioning holes in the openable and closable exterior covers and providing the projections on the left cover **505(a)**.

(87) A second exemplary embodiment will be described. The configuration related to image formation in the second exemplary embodiment is the same as that of the image forming apparatus **101** in the first exemplary embodiment, and thus description thereof will be omitted.

(88) The toner bottle door **501** and the collection container door **503** in the first exemplary embodiment each include the hinge configuration at the lower end of the door, and are opened and closed in the up-down direction around the rotation center at the lower end of the door as described in FIGS. **5A** and **5B**, and the projections **551** and **553** provided on the toner bottle door **501** and the collection container door **503** fit into the positioning holes **601(a)** and **601(c)** provided in the left cover **505(a)**, respectively, so that the doors are positioned.

(89) However, it is presumed that there may be no space for providing the positioning holes in the left cover **505(a)** depending on the configuration of the image forming unit **102**.

(90) As illustrated in FIG. **11**, in a hinge part including a hinge shaft **703** that fits into the rotating part of a hinge plate **701** of a toner bottle door **501** and to a hinge base **702** on the main body side, a compression spring **704** that tends to extend in both directions shown by arrow **11** is interposed to bias the hinge plate **701** in the direction of arrow **12** in FIG. **11**. That is, the toner bottle door **501** itself is biased toward the left side of the main body (biased in the direction of arrow **13** in FIG. **12**) so that thrust play is shifted in the axial direction of the hinge part. On the other hand, as illustrated

in FIG. 12, a left cover 505(b) is provided with an abutment part 602 instead of a positioning hole. (91) According to the configuration described above, the toner bottle door 501 is always biased in the left direction of the main body. When the door is being closed, the side surface of a projection 551 abuts the abutment part 602 of the left cover 505(b) so that the toner bottle door 501 itself is positioned in the state of being shifted to the left. (92) The above description has been made taking the toner bottle door 501 as an example. A collection container door 503 can be similarly positioned by providing the same configuration. (93) In addition, the above description has been made taking the cover member on the left side of the main body as an example. The same advantageous effect can be achieved by providing the same configuration on the right side. (94) Further, the same advantageous effect can be achieved by providing the projections on the covers and the abutment parts on the doors. (95) The same advantageous effect can be achieved by selectively choosing for each door the positioning configuration in the first exemplary embodiment and the configuration in the second exemplary embodiment. (96) According to the present disclosure, in an image forming apparatus including a plurality of doors, it is possible to reduce misalignment between relative positions of adjacent doors in the horizontal direction. (97) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions. (98) This application claims the benefit of Japanese Patent Application No. 2023-065263, filed Apr. 12, 2023, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus including an image forming unit configured to form an image on a sheet, the image forming apparatus comprising: a housing configured to store the image forming unit; a plurality of doors, each of which rotates around a rotational axis so as to be opened and closed with respect to the housing and includes a positioning shaft and a projection, wherein, when each of the plurality of doors is in an opened state, an inside of the image forming apparatus is exposed; a cover part that has a plurality of fitting holes into which the positioning shafts are fitted; a plurality of holding members, each of which holds a different one of the plurality of doors in a closed state; a sheet storage part configured to store the sheet; and a sheet cassette cover attached to the sheet storage part wherein, when each of the plurality of doors is in the closed state, each of the positioning shafts included in the plurality of doors is fitted into a different one of the plurality of fitting holes, wherein the plurality of doors and the sheet cassette cover are arranged in a vertical direction, and wherein the sheet cassette cover is configured to be adjustable in a horizontal direction with respect to the sheet storage part for alignment with the plurality of doors.
2. The image forming apparatus according to claim 1, wherein, among the plurality of doors, two doors adjacent to each other in the vertical direction are identical in width in the horizontal direction, and when the two doors adjacent to each other are in the closed state, end portions of the two doors in the horizontal direction are located at an identical position in the horizontal direction.
3. The image forming apparatus according to claim 1, wherein a first door that is at least one door of the plurality of doors rotates around a rotational axis extending in the vertical direction, wherein a second door that is at least one door of the plurality of doors rotates a rotational axis extending in the horizontal direction, and wherein the first door and the second door are adjacent to each other in the vertical direction.
4. The image forming apparatus according to claim 1, wherein each of the plurality of doors

includes the projection, and wherein the image forming apparatus further comprises a plurality of sensors, each of which detects that a different one of the plurality of doors is in the closed state by detecting a different one of the plurality of projections included in the plurality of doors.

5. The image forming apparatus according to claim 1, wherein the plurality of fitting holes is provided along the vertical direction.

6. The image forming apparatus according to claim 1, wherein each of the positioning shafts included in the plurality of doors protrudes in a direction away from a different one of the plurality of doors, and each of cross section areas of the positioning shafts is reduced as a distance from the different one of the plurality of doors increases.

7. The image forming apparatus according to claim 1, wherein, when one of the positioning shafts included in a door of the plurality of doors is fitted into a different one fitting hole of the plurality of fitting holes, where the different one fitting hole restricts a position of the door provided with the fitted positioning shaft in the horizontal direction.

8. The image forming apparatus according to claim 7, wherein, when one of the positioning shafts included in a door the plurality of doors is fitted into a different one fitting hole of the plurality of fitting holes, the one different fitting hole restricts a position of the door provided with the fitted positioning shaft in the vertical direction.

9. The image forming apparatus according to claim 1, further comprising a plurality of metallic parts, wherein each of the plurality of doors includes a magnet, and wherein, when each of the plurality of doors is in the closed state, each of the plurality of magnets included in the plurality of doors attracts a different one of the plurality of metallic parts.

10. The image forming apparatus according to claim 1, wherein, among the plurality of doors, two doors adjacent to each other are provided side by side in the vertical direction, and wherein the two doors adjacent to each other are identical in length in the horizontal direction, and when the two doors are in the closed state, a difference between relative positions of end portions of the two doors in the horizontal direction is less than 2 millimeters (mm).

11. The image forming apparatus according to claim 1, further comprising: a plurality of sensors, each of which detects that a different one of the plurality of doors is in the closed state by detecting a different one of the plurality of projections included in the plurality of doors, wherein, when each of the plurality of doors is in the opened state, the image forming unit is accessible, and wherein, among the plurality of doors, two doors adjacent to each other in the vertical direction are identical in width in a horizontal direction, and when the two doors adjacent to each other are in the closed state, end portions of the two doors in the horizontal direction are located at an identical position in the horizontal direction.

12. The image forming apparatus according to claim 1, wherein each of the plurality of fitting holes is elongated in the vertical direction for positioning the plurality of doors in the horizontal direction.

13. The image forming apparatus according to claim 12, wherein the plurality of fitting holes is provided along the vertical direction.

14. The image forming apparatus according to claim 1, wherein the plurality of doors at least includes two adjacent doors which are arranged in the vertical direction to be adjacent to each other, wherein a first door of the two adjacent doors has a first door length in the vertical direction, and a second door of the two adjacent doors has a second door length in the vertical direction, and wherein the cover part has a cover length in the vertical direction which is longer than at least a sum of the first door length and the second door length.
